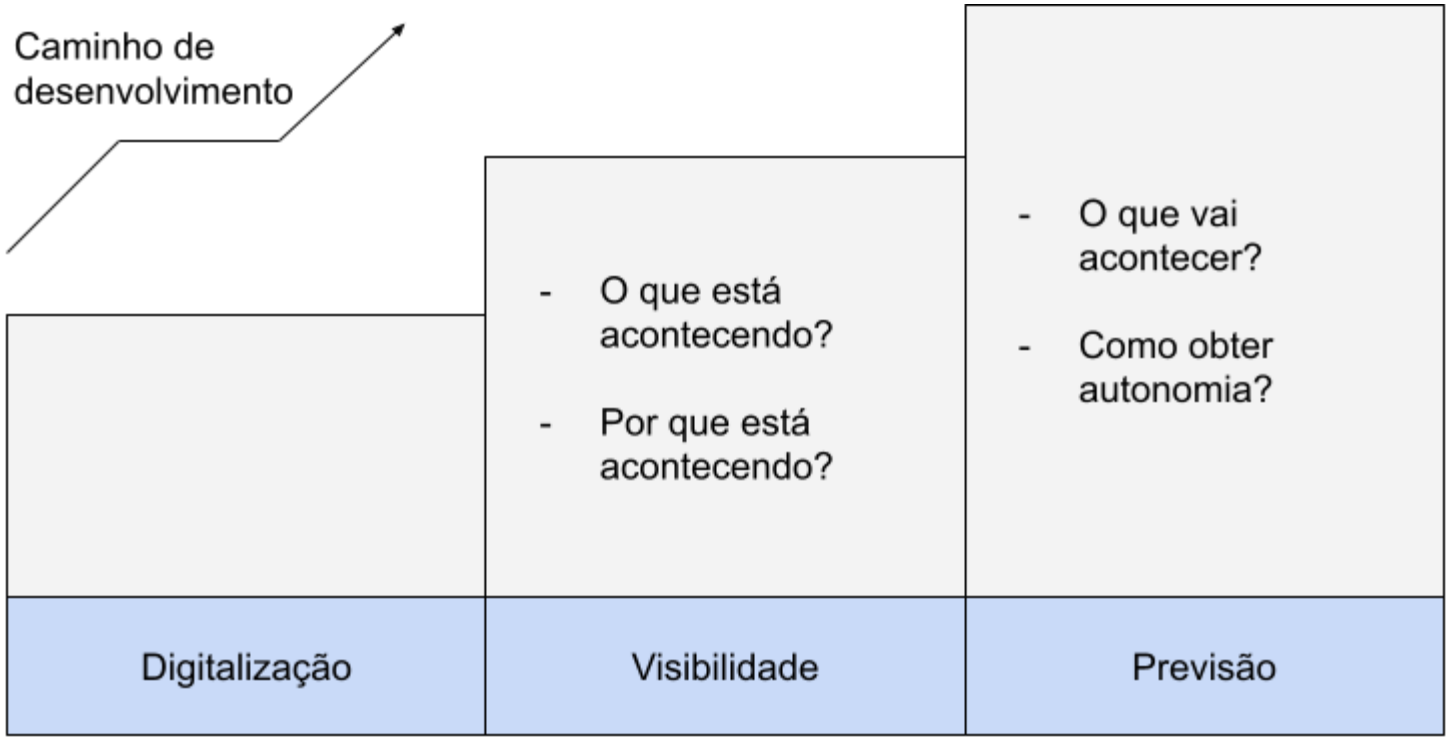
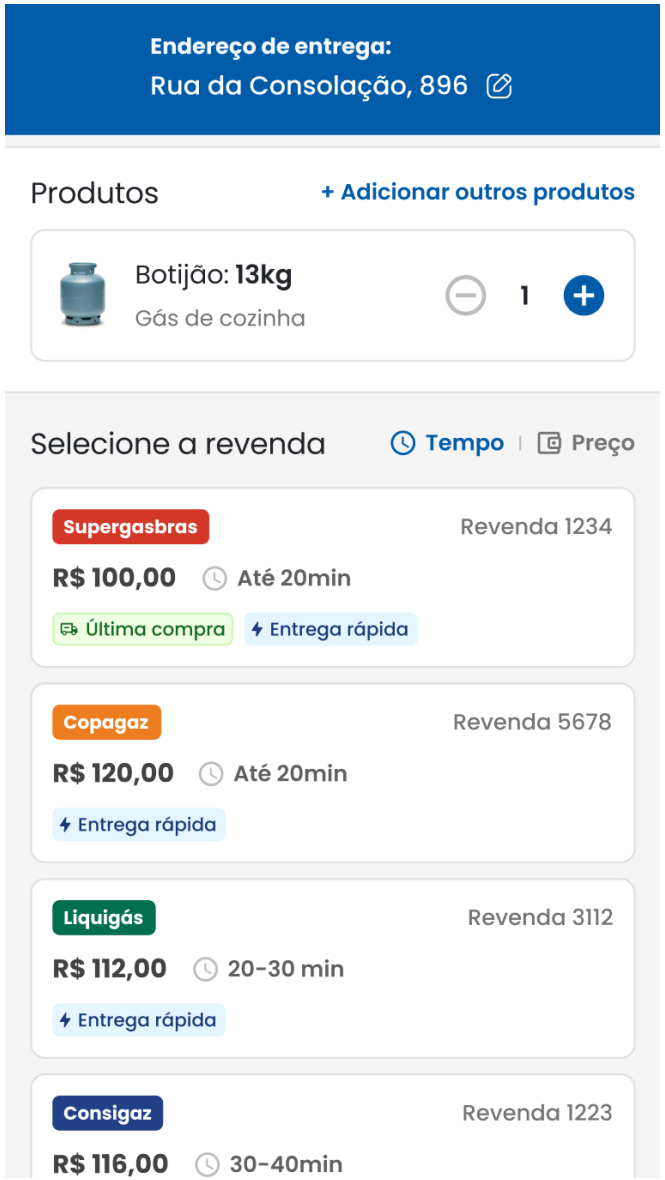
Previsão e Controle de Tempos das Entregas em Plataformas de Serviços com Inteligência Artificial

Guilherme Barrado Pereira, Rogério de Oliveira

Mackenzie

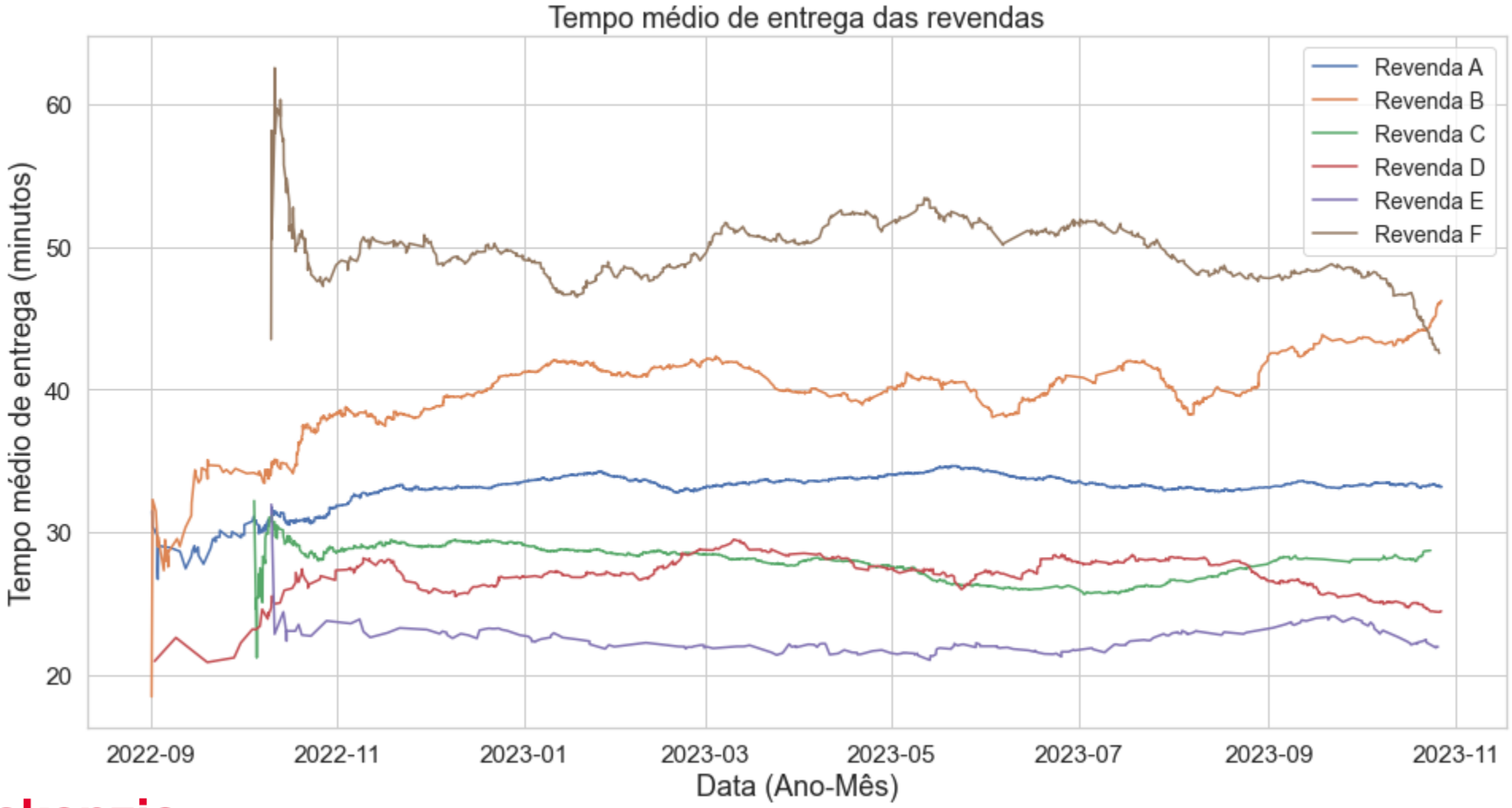
Introdução





Mackenzie

Introdução



Mackenzie

Referencial teórico

- Decisões orientadas a dados
  - Aumento de competitividade
  - Carros autônomos, Reconhecimento de imagem, Sistemas de recomendação

- Aprendizado de máquina supervisionado
  - Decision Trees, Random Trees, Support Vector Machines

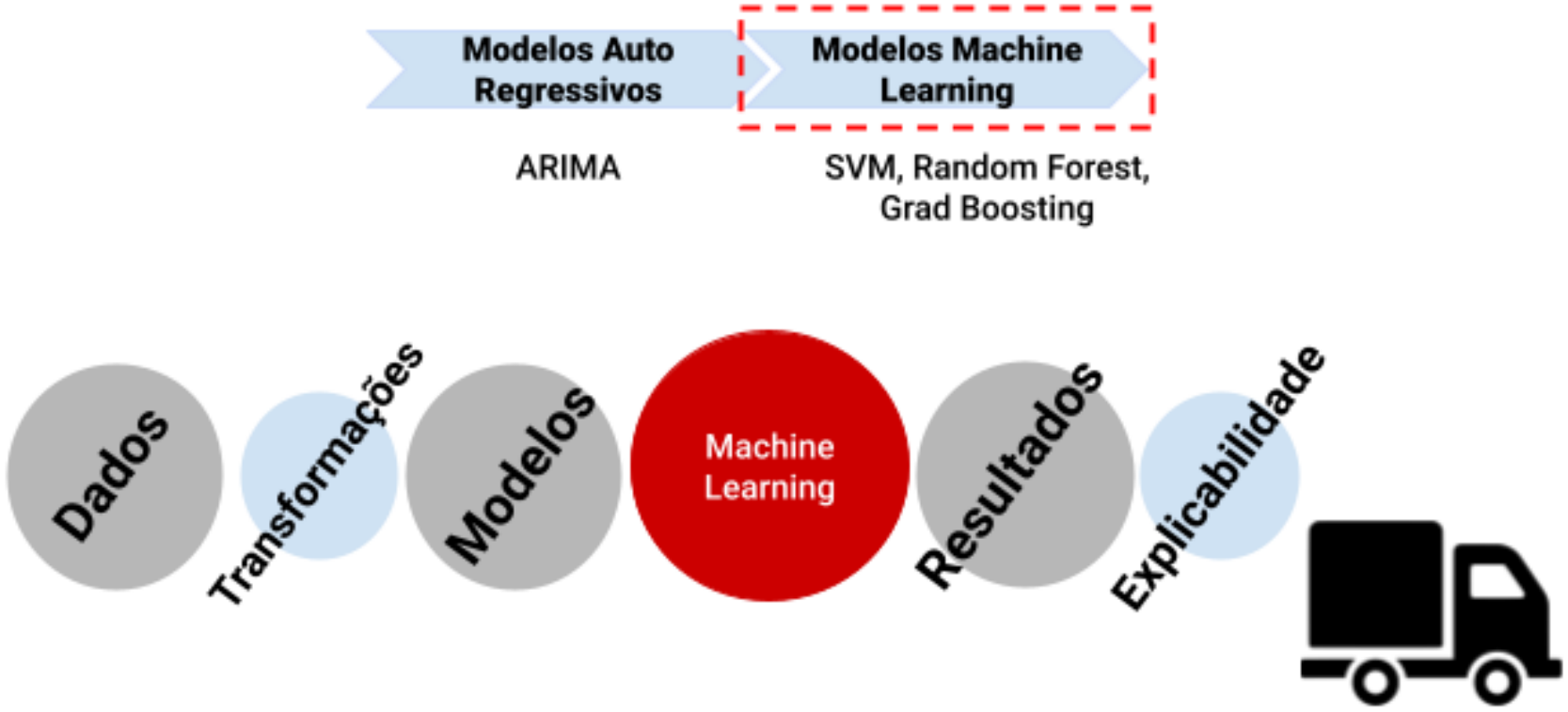
Mackenzie

Referencial teórico

- Trabalhos correlatos:
  - Previsão de tempo de viagem em transporte de passageiros (Taxi)
    - Random Forest, SVM, Extremely Randomized Trees
  - Previsão de tempo de viagem em transporte multimodal de cargas
    - SVM, AdaBoost, Extremely Randomized Trees

Mackenzie

Metodologia



Mackenzie

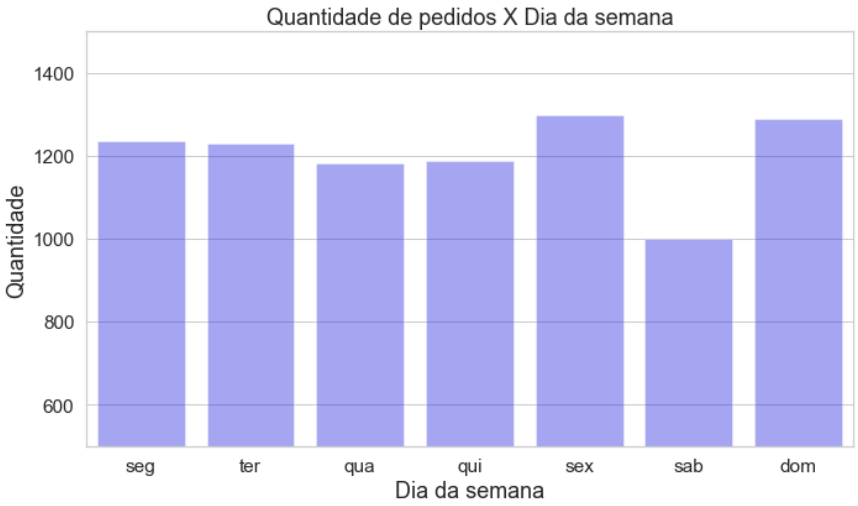
Metodologia - Dados

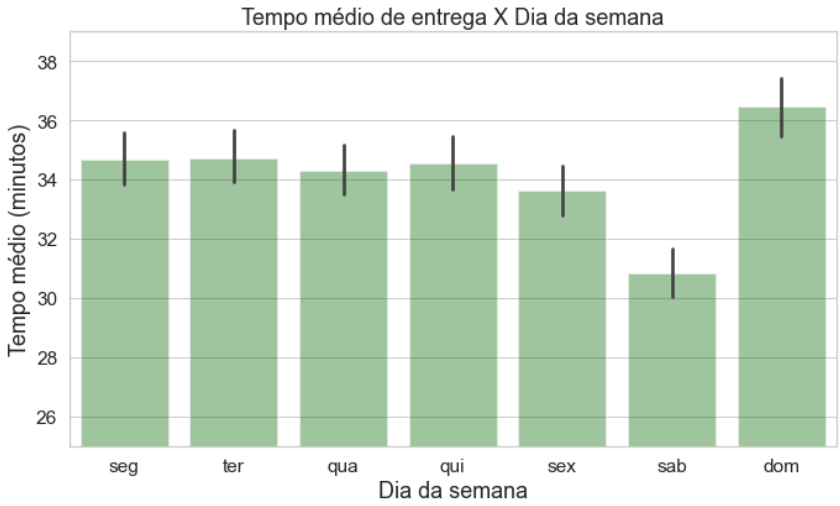
- Conjunto total: 8.421
- Outliers: 358
- Período: set/22 a nov/2023

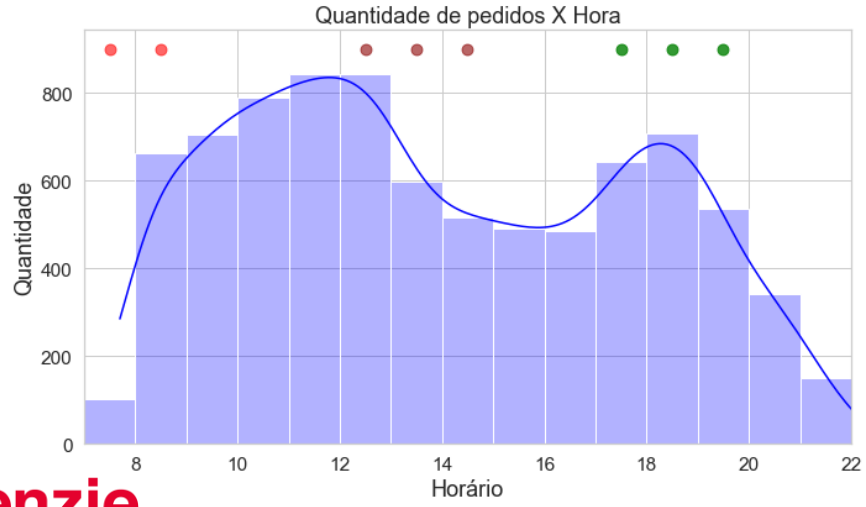
quantity	holiday	time_elapsed	ongoing_orders	waze_avg_time	waze_avg_distance	datetime	reseller
1	False	18.459633	1	8.583333	3.136	2022-09-01 10:50:00	B
1	False	31.472517	0	11.183333	5.397	2022-09-01 13:21:00	A
1	False	29.343333	0	8.400000	3.123	2022-09-01 17:51:00	A
1	False	46.103800	0	12.733333	4.536	2022-09-01 18:35:00	B
1	False	20.950450	0	8.866667	3.524	2022-09-02 10:13:00	D

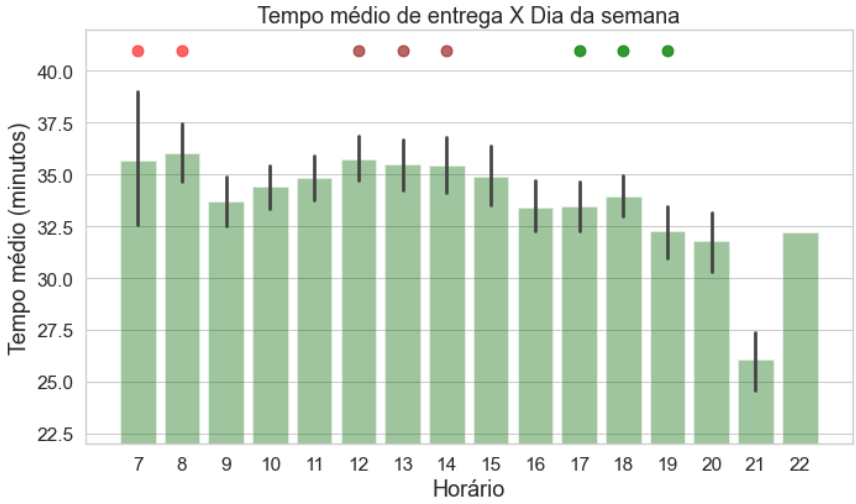
Mackenzie

Metodologia – Análise exploratória





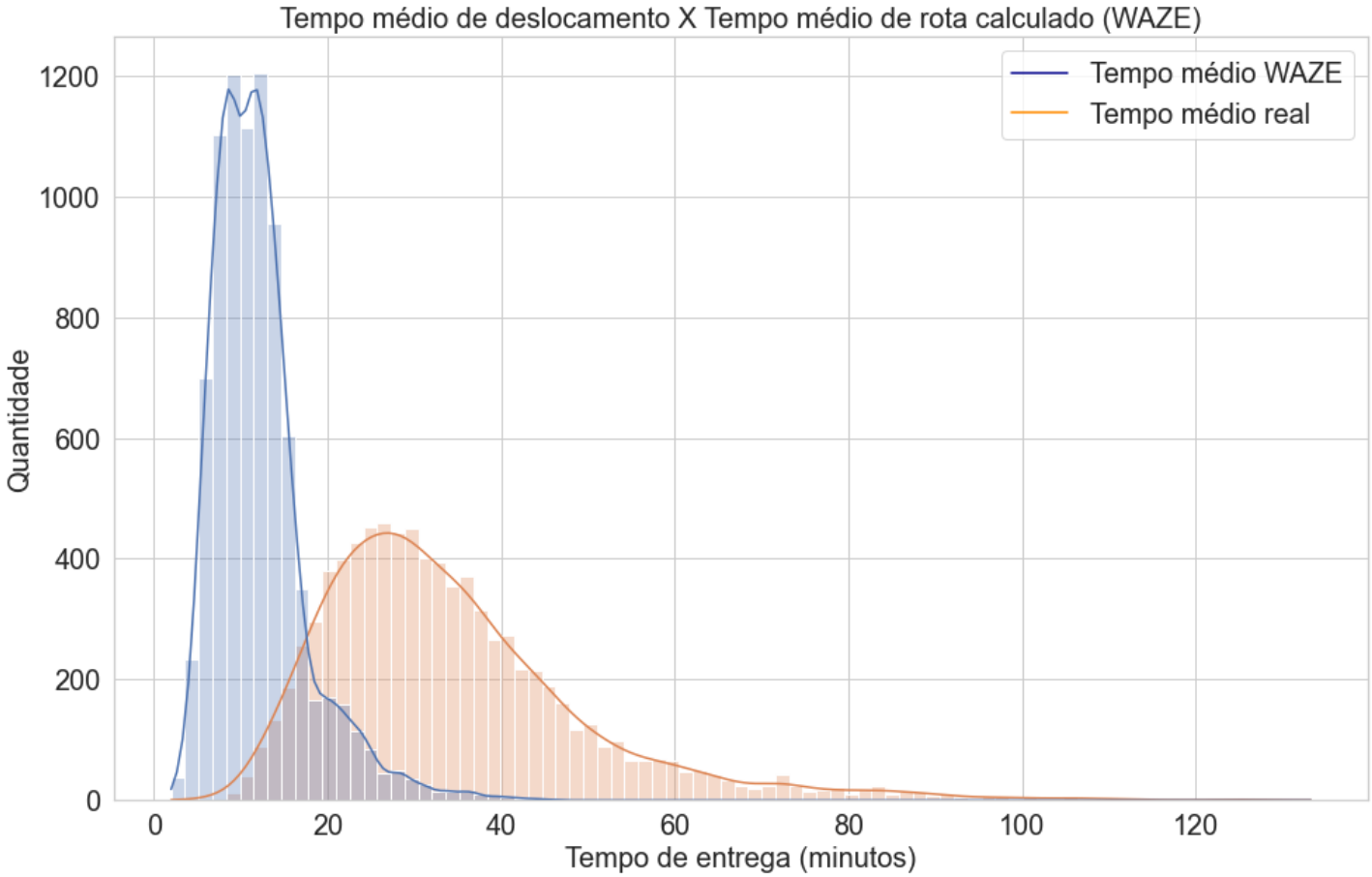




Mackenzie

Metodologia – Preparação dos dados

- Novas variáveis
  - Distância (Waze)
  - Tempo de rota (Waze)



Mackenzie

Metodologia – Preparação dos dados

- Variáveis derivadas:
  - Hora em formato decimal
  - Dia do mês
  - Dia da semana

- Encoding e Normalização (Escala)
  - Holiday
  - Reseller

hour	day	weekday
10.83	1	3
13.35	1	3
17.85	1	3
18.58	1	3
10.22	2	4

Mackenzie

Metodologia – Treinamento e Seleção

- Treinamento com todos os dados
- Seleção dos 3 melhores

Regressor	mae	mse	mape
extra trees	5.007639	37.657613	0.162562
rfe	5.025565	37.782627	0.163192
random forest	5.028963	37.813240	0.163287
svm	5.137934	40.689942	0.164663
decision tree	5.088653	39.147300	0.165164
grad boosting	5.061172	38.246626	0.165454
bagging	5.140261	39.601932	0.167118
hist grad boosting	5.084300	38.649263	0.167784
ada boost	5.210779	39.713190	0.174214

Mackenzie

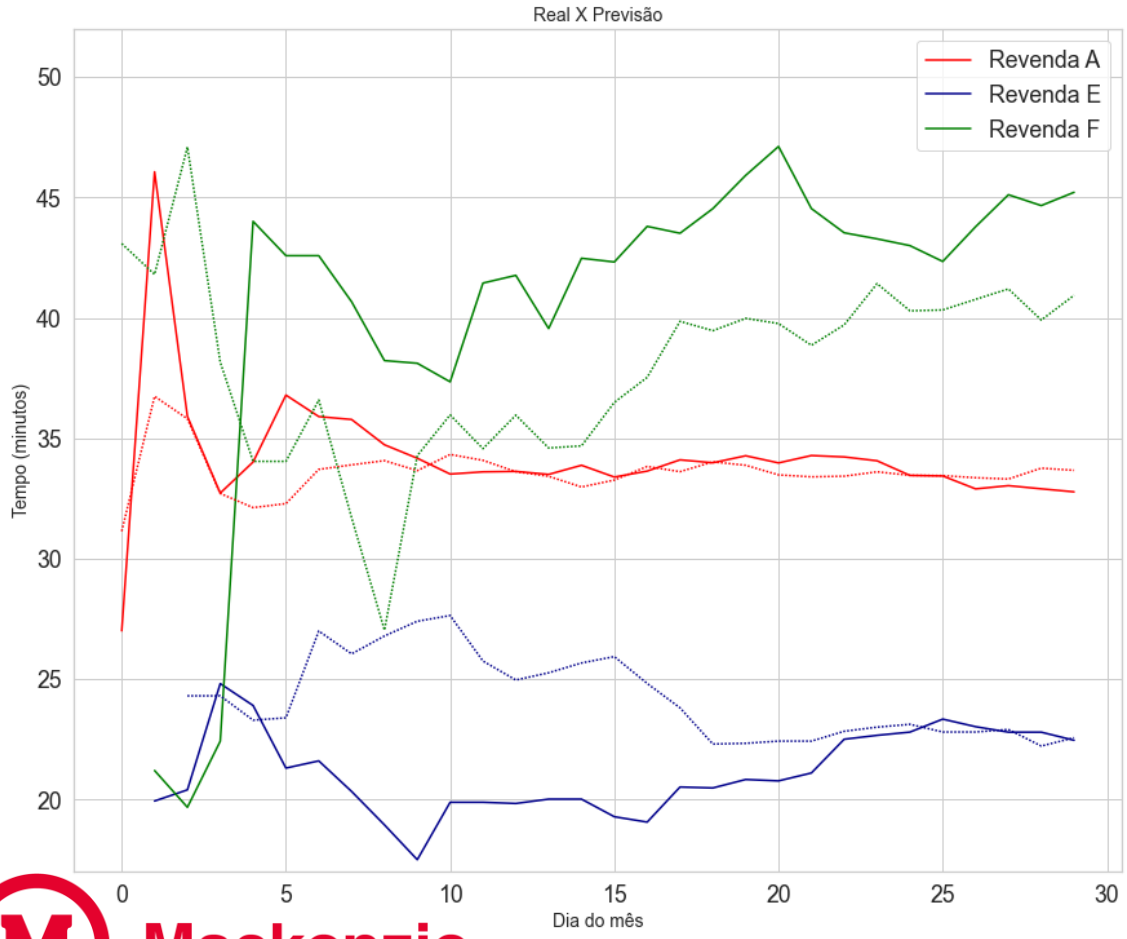
Resultados - Previsão

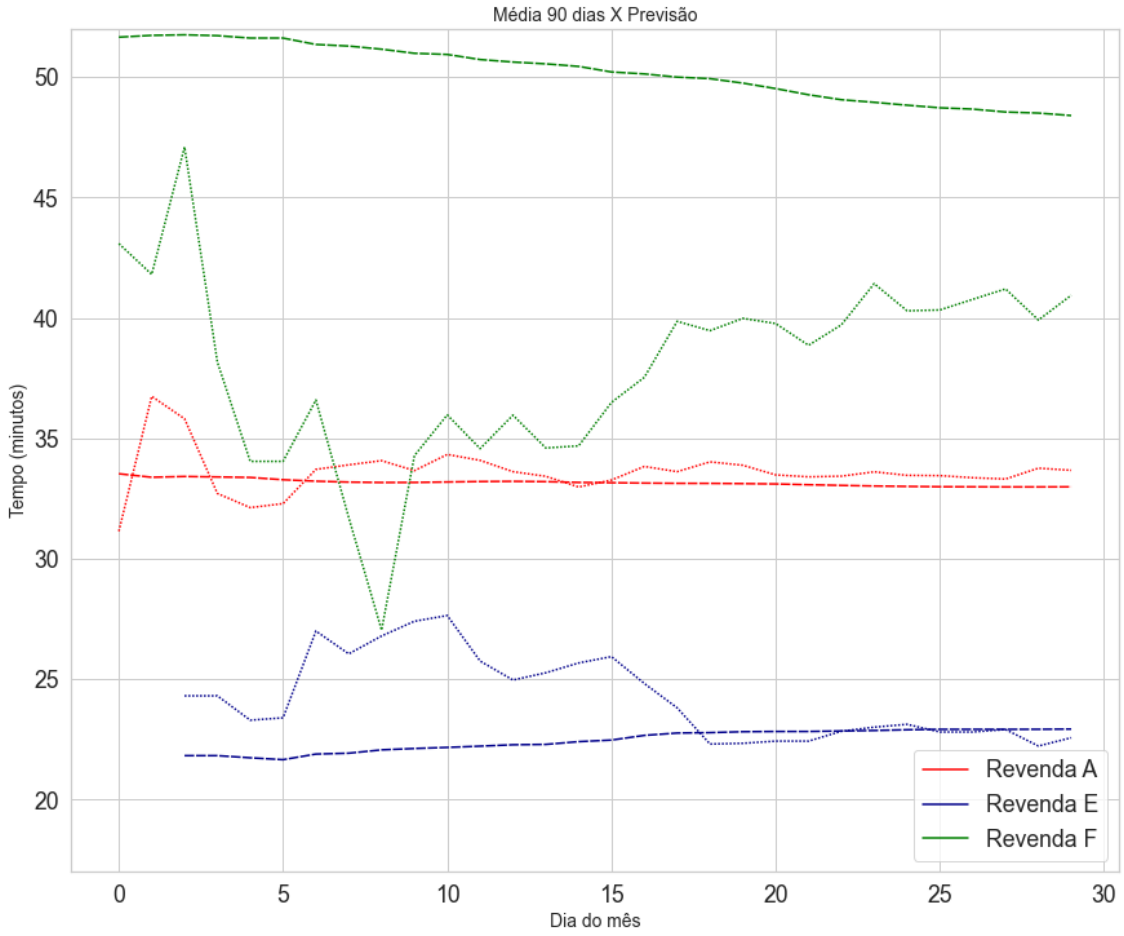
- Período de treinamento: 90 dias
- Período previsto: 30 dias

Regressor	M.A.P.E.	M.A.E.
Extra Trees	0.164471	4.376498
Random Forest	0.161812	4.314617
Gradient Boosting	0.164357	4.290353
Método atual	0.202354	4.572104

Mackenzie

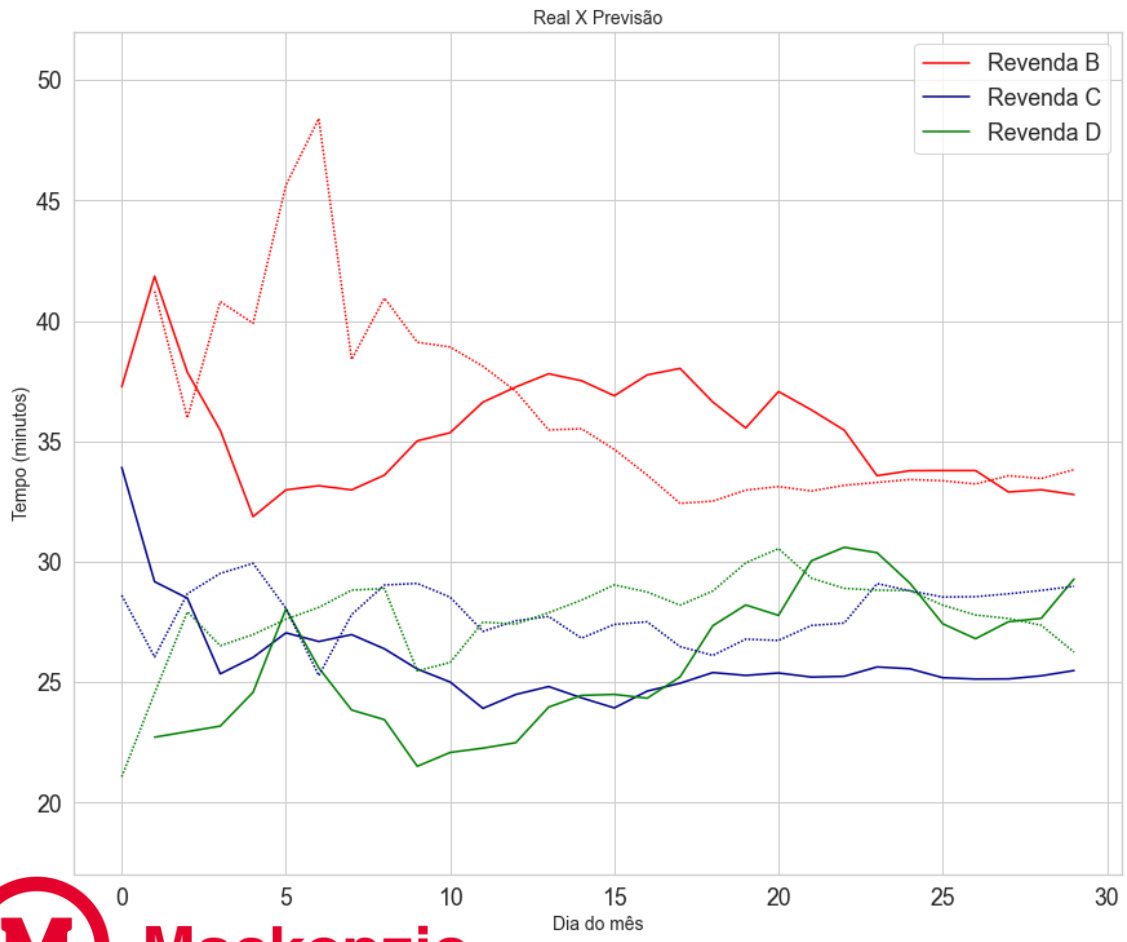
Resultados - Previsão

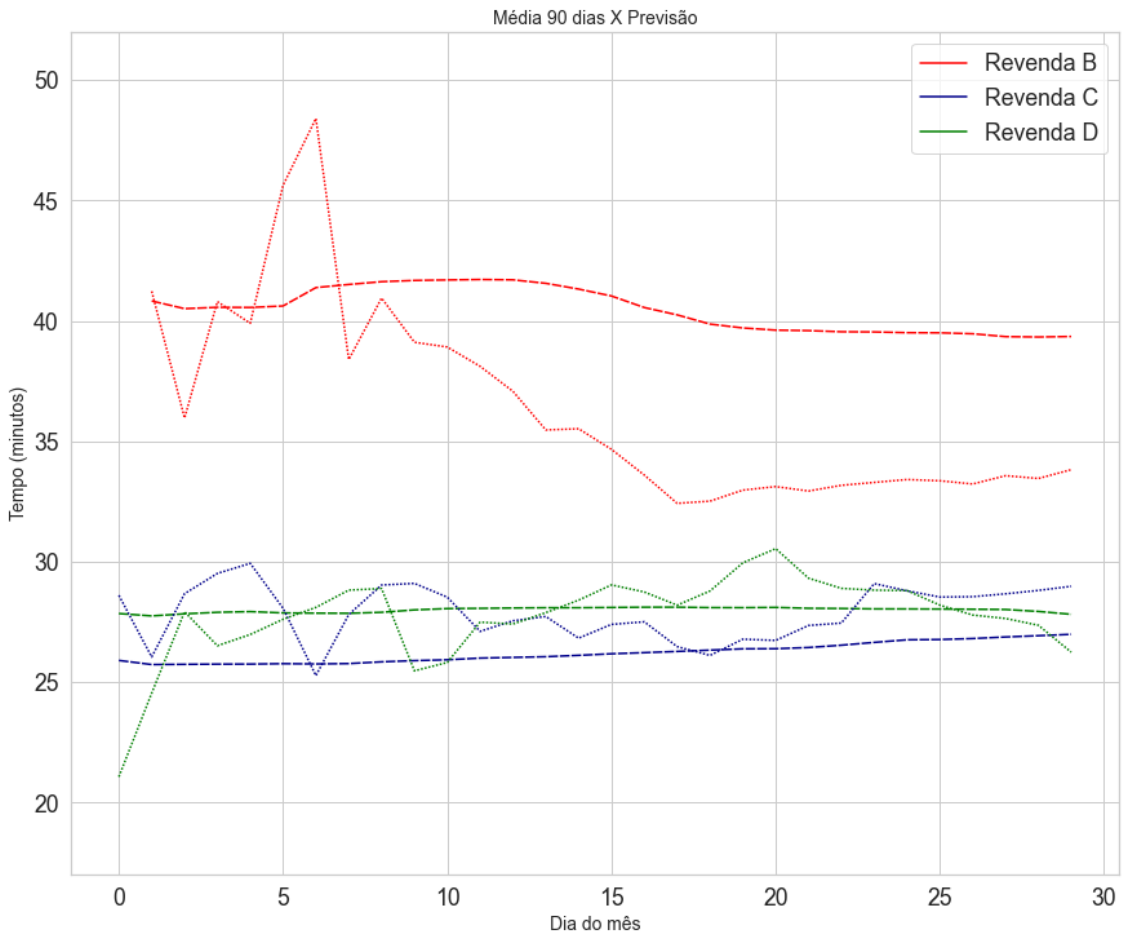




Mackenzie

Resultados - Previsão





Mackenzie

Resultados - Explicabilidade

- Variáveis mais impactantes:
  - Tempo médio da rota entre endereços
  - Distância média da rota entre endereços
  - Revendedor

Features	Importances
waze_avg_time	0.763503
waze_avg_distance	0.178882
reseller_label	0.020250
ongoing_orders	0.014076
hour	0.006652
weekday	0.006543
day	0.006405
quantity	0.001955
holiday_label	0.001734

Mackenzie

Conclusão

- Extremely Randomized Trees (*MAPE* 16%) tem desempenho melhor do que o utilizado atualmente pela empresa (*MAPE* 20%)

- Erro Absoluto Médio (*MAE*) de 4.37 minutos

Mackenzie

Conclusão

- Trabalhos futuros
  - Novas variáveis
  - Aplicação em outras plataformas
  - Outras técnicas de Aprendizado de máquina

Mackenzie

Referências

Almaghrebi, A., Aljuheshi, F., Rafaie, M., James, K. e Alahmad, M. (2020) “Data-Driven Charging Demand Prediction at Public Charging Stations Using Supervised Machine Learning Regression Methods”, Energies, vol. 13, no. 16.

Bughin, J., Seong, J., Manyika, J., Chui, M. e Joshi, R. (2018), “Notes from the AI Frontier Modeling the Impact of AI On The World Economy”, McKinsey Global Institute.

Henke, N., Bughin, J., Chui, M., Manyika, J., Saleh, T., Wiseman, B. e Sethupathy, G. (2016), “The Age Of Analytics: Competing In A Data-Driven World”, McKinsey Global Institute.

Joshi, N., Hotalappa, M. e Gadade, K. (2017) “A Study on Travel Time Estimation for Taxi Trips”, International Education & Research Journal (IERJ), vol. 3.

Mackenzie

Referências

Lam, H. T., Diaz-Aviles, E., Pascale, A., Gkoufas, Y. e Chen, B. (2015) “Taxi Destination and Trip Time Prediction from Partial Trajectories”, arXiv.

Moreira, J. M., Jorge, A., Sousa, J. F. e Soares, C. (2005) “Trip Time Prediction In Mass Transit Companies. A Machine Learning Approach”, Advanced OR and AI methods in transportation.

Servos, N., Liu, X., Teucke, M. e Freitag, M. (2020) “Travel Time Prediction in a Multimodal Freight Transport Relation Using Machine Learning Algorithms”, Logistics 2020

Shankar, A., Jebarajakirthy, C., Nayal, P., Maseeh, H. I., Kumar, A., & Sivapalan, A. (2022). Online food delivery: A systematic synthesis of literature and a framework development. International Journal of Hospitality Management, 104, 103240.

Mackenzie